

# 1.3 Hazardous Earth

This topic covers tectonic hazards (earthquakes, volcanoes, tsunamis), atmospheric hazards (tropical storms), and wildfire. It examines the physical processes that create hazards, the factors that influence their severity and impacts, and how societies manage risk through prediction, protection, and planning.

## Plate Tectonics Theory

**Plate tectonics:** The theory that Earth's lithosphere is divided into large rigid plates that move relative to one another, driven by convection currents in the mantle and gravitational forces at subduction zones. Tectonic hazards (earthquakes, volcanoes) are concentrated at plate boundaries, though intraplate hazards also occur (e.g., the New Madrid Seismic Zone, USA).

## Earth's Internal Structure

| Layer                        | Composition                        | State                                                                 | Depth          |
|------------------------------|------------------------------------|-----------------------------------------------------------------------|----------------|
| Inner core                   | Iron & nickel                      | Solid (immense pressure keeps it solid despite ~5,400°C temperatures) | 5,150–6,371 km |
| Outer core                   | Iron & nickel                      | Liquid; convection here generates Earth's magnetic field              | 2,900–5,150 km |
| Lower mantle                 | Dense silicate rock (perovskite)   | Solid but capable of slow flow over geological timescales             | 660–2,900 km   |
| Upper mantle (asthenosphere) | Silicate rocks (olivine, pyroxene) | Semi-molten, ductile; plates float on this layer                      | 100–660 km     |

|                     |                                                        |                                             |         |
|---------------------|--------------------------------------------------------|---------------------------------------------|---------|
| Crust (oceanic)     | Basalt (mafic);<br>density $\sim 3.0 \text{ g/cm}^3$   | Solid; thin (5–10 km); younger (<200 Ma)    | 0–10 km |
| Crust (continental) | Granite (felsic);<br>density $\sim 2.7 \text{ g/cm}^3$ | Solid; thick (25–70 km); older (up to 4 Ga) | 0–70 km |

**Lithosphere:** The rigid outer layer comprising the crust and uppermost mantle (~100–250 km thick). It is broken into 7 major and several minor tectonic plates.

**Asthenosphere:** The semi-molten, ductile layer beneath the lithosphere (approximately 100–660 km depth) on which lithospheric plates move. Partial melting here produces magma that feeds volcanic activity.

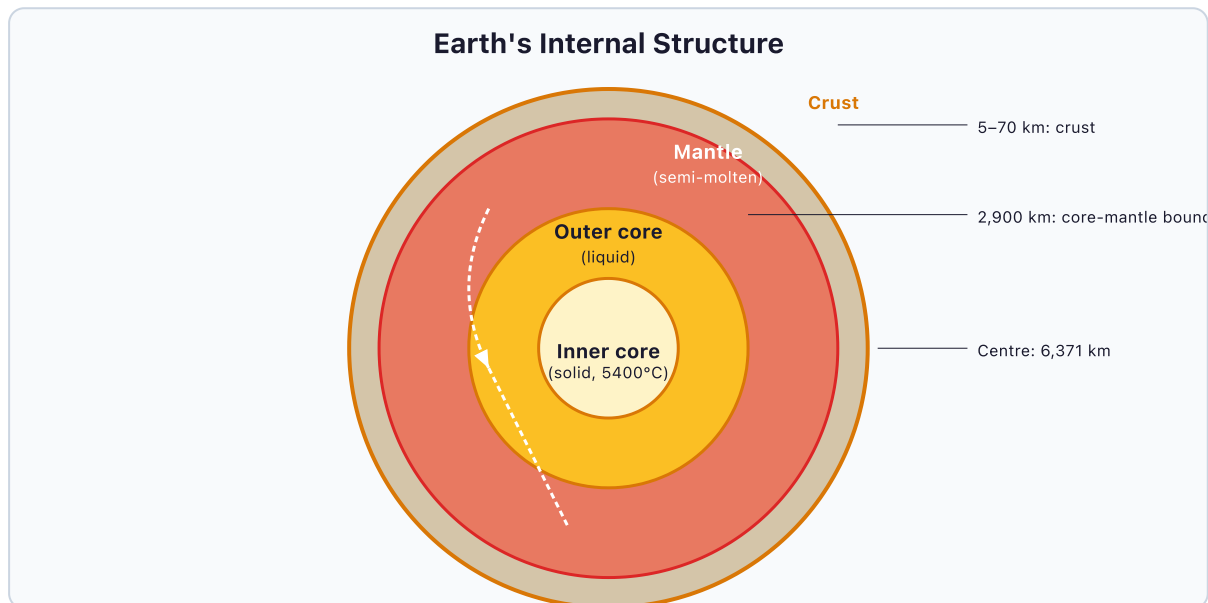
## Evidence for Plate Tectonics

- **Continental fit:** Alfred Wegener (1912) noted the jigsaw fit of continents (e.g., eastern South America and western Africa coastlines match closely)
- **Fossil evidence:** Identical fossils of Mesosaurus (freshwater reptile) and Glossopteris (fern) found on now-separated continents — organisms that could not have crossed an ocean
- **Rock and geological matches:** The Appalachian Mountains (USA) and Caledonian Mountains (Scotland/Norway) are geologically continuous despite being separated by the Atlantic
- **Palaeomagnetism:** Alternating normal and reversed magnetic stripes in ocean-floor basalt, symmetrical either side of mid-ocean ridges, proving sea-floor spreading (Vine and Matthews, 1963)
- **GPS measurements:** Direct satellite measurement confirms plate movement of 1–15 cm/year (e.g., North Atlantic is widening by  $\sim 2.5 \text{ cm/year}$ )
- **Hot spots:** Chains of volcanic islands (e.g., Hawaiian chain) show progressive aging away from the active volcano, indicating plate movement over a stationary mantle plume
- **Ocean floor age:** Youngest oceanic crust is found at mid-ocean ridges, with age increasing symmetrically towards continents

## Driving Mechanisms

- **Mantle convection:** Radioactive decay in the core and mantle generates heat, driving convection currents. Rising material at constructive boundaries pushes plates apart; sinking material at destructive boundaries pulls them together

- **Ridge push:** Newly formed hot crust at mid-ocean ridges stands elevated; gravity pushes the plate downhill away from the ridge (gravitational sliding)
- **Slab pull:** Dense, cold oceanic lithosphere subducts and sinks into the mantle under gravity, pulling the rest of the plate behind it. This is considered the dominant mechanism — plates with large subducting slabs (e.g., Pacific Plate) move fastest
- **Basal drag:** Friction between the base of plates and underlying mantle convection currents may contribute to plate motion, though its importance is debated



## Plate Boundaries

### Divergent (Constructive) Boundary

Plates move apart due to rising convection currents and ridge push. Magma rises from the asthenosphere to fill the gap, creating new oceanic crust through sea-floor spreading. Typically 2–5 cm/year separation rate. Features include mid-ocean ridges (underwater mountain chains), rift valleys, shield volcanoes, frequent but shallow earthquakes (usually <M6), and high geothermal heat flow.

- **Oceanic-oceanic divergence:** Mid-Atlantic Ridge — 16,000 km long, rising above sea level in Iceland. The ridge produces frequent volcanic eruptions (e.g., Eyjafjallajökull, 2010) and earthquakes along transform faults
- **Continental-continental divergence:** East African Rift System — 3,000 km from the Afar Triangle (Ethiopia) to Mozambique. The African Plate is splitting into the Nubian and Somali plates at ~7

mm/year. Features alkaline lakes, volcanoes (Kilimanjaro – dormant; Nyiragongo – active), and deep rift valleys. In 50–100 million years, East Africa may separate completely, forming a new ocean

## Convergent (Destructive) Boundary

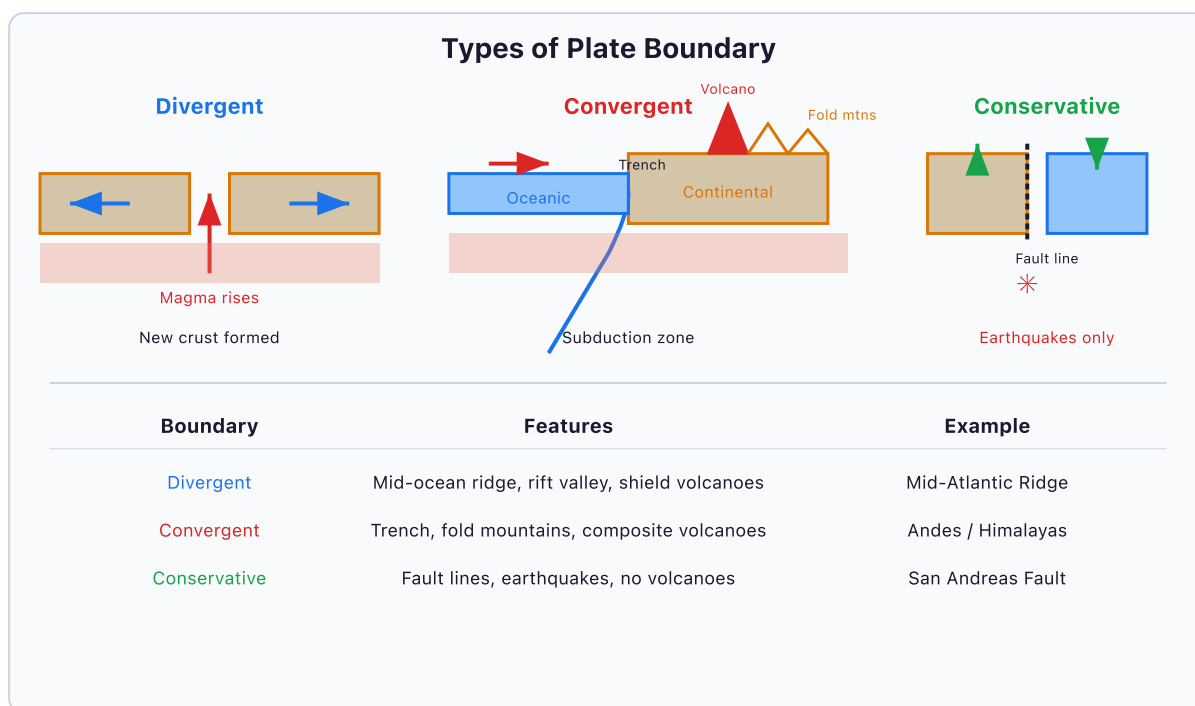
Plates move together. Where oceanic crust meets continental or other oceanic crust, the denser plate subducts (descends) into the asthenosphere, where it melts. Subduction produces: deep ocean trenches, fold mountains, explosive composite volcanoes (from partial melting of the subducting slab and overlying mantle wedge), and earthquakes at all depths (shallow to 700 km along the Wadati-Benioff zone).

- **Oceanic-continental:** Nazca Plate subducting beneath South American Plate at ~6 cm/year → Peru-Chile Trench (8,065 m deep), Andes mountains (6,961 m), explosive volcanism (e.g., Cotopaxi, Nevado del Ruiz). The angle and rate of subduction influence eruption style and earthquake depth
- **Oceanic-oceanic:** Pacific Plate subducting beneath Philippine Plate → Mariana Trench (10,994 m, deepest point on Earth), island arc volcanoes (Mariana Islands). Produces intense deep-focus earthquakes
- **Continental-continental:** Indian Plate colliding with Eurasian Plate at ~4–5 cm/year → Himalayas (rising ~5 mm/year; Everest 8,849 m). No subduction occurs because both plates have similar low density; instead, crust crumples, folds, and thickens. No volcanic activity but intense earthquakes (e.g., Nepal 2015)

## Conservative (Transform) Boundary

Plates slide past each other laterally. No crust is created or destroyed. Friction causes plates to lock along the fault plane; strain energy builds until it is released suddenly as an earthquake. No volcanic activity occurs. Movement is often jerky rather than smooth, producing a stick-slip mechanism.

Example: San Andreas Fault, California — Pacific Plate moves NW relative to North American Plate at ~46 mm/year. The locked southern section has not ruptured since 1857 and is considered overdue for a major earthquake (the "Big One").



## Volcanic Hazards

**Volcano:** An opening in the Earth's crust through which magma, ash, and gases are erupted. Volcanoes form at constructive boundaries (basaltic, effusive), destructive boundaries (andesitic/rhyolitic, explosive), and over hot spots (e.g., Hawaii). There are approximately 1,500 potentially active volcanoes worldwide, with 50–70 erupting each year.

## Types of Volcano

| Type   | Shape                                         | Eruption Style                                               | Lava Type                                              | Example                                                                                                 |
|--------|-----------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Shield | Wide, gently sloping (slope <10°); very large | Effusive: gentle lava flows, lava fountains; relatively safe | Basaltic (low silica ~50%, low viscosity, 1000–1200°C) | Mauna Loa, Hawaii (largest active volcano on Earth; 4,169 m above sea level, ~9,100 m from ocean floor) |

|                              |                                                                                                                          |                                                                                          |                                                                          |                                                                                              |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Composite<br>(stratovolcano) | Steep,<br>symmetrical cone;<br>built of alternating<br>lava and ash<br>layers                                            | Explosive:<br>pyroclastic flows,<br>ash columns,<br>lahars; highly<br>dangerous          | Andesitic/rhyolitic<br>(high silica 60–<br>75%, viscous, 800–<br>1000°C) | Mount St Helens,<br>USA (1980<br>eruption: VEI 5)                                            |
| Caldera                      | Massive<br>depression (up to<br>80 km diameter)<br>from collapsed<br>magma chamber<br>after<br>supervolcanic<br>eruption | Supervolcanic:<br>catastrophically<br>explosive (VEI 7–<br>8); global climate<br>impacts | Rhyolitic (very high<br>silica >70%,<br>extremely viscous)               | Yellowstone,<br>USA (last super-<br>eruption<br>640,000 years<br>ago; caldera 55<br>× 72 km) |
| Cinder cone                  | Small, steep-<br>sided (slope 30–<br>40°); usually <300<br>m tall                                                        | Short-lived<br>eruptions; tephra<br>and lava bombs                                       | Basaltic to andesitic                                                    | Paricutin, Mexico<br>(grew in a<br>farmer's<br>cornfield, 1943–<br>1952)                     |

## Volcanic Hazards in Detail

- **Lava flows:** Molten rock flowing downslope at 1–30 km/h (basaltic) or creeping at metres per hour (viscous lava). Destroys everything in its path but usually slow enough for evacuation. Basaltic lava from shield volcanoes travels much further than viscous andesitic/rhyolitic lava
- **Pyroclastic flows:** Fast-moving (100–700 km/h) ground-hugging currents of superheated gas (up to 1,000°C), ash, and rock fragments. The deadliest volcanic hazard; almost impossible to outrun. Killed ~29,000 at Mount Pelee, Martinique (1902)
- **Tephra and ash fall:** Fragmented material ejected into the atmosphere. Fine ash can collapse roofs (at 10+ cm depth), contaminate water supplies, destroy crops, disrupt aviation (jet engines ingest ash), and cause respiratory illness. The 2010 Eyjafjallajökull eruption in Iceland disrupted 100,000 flights and cost airlines \$1.7 billion
- **Lahars:** Volcanic mudflows created when volcanic material mixes with water (rainfall, melted glacial ice, breached crater lakes). Can travel at 60+ km/h and reach 100+ km from the volcano. The

Nevado del Ruiz eruption (1985, Colombia) generated lahars that buried the town of Armero, killing 23,000 people

- **Volcanic gases:** CO<sub>2</sub>, SO<sub>2</sub>, H<sub>2</sub>S, HCl, HF released during eruptions. Can be toxic (CO<sub>2</sub> is denser than air and can accumulate in depressions — Lake Nyos disaster, Cameroon, 1986: CO<sub>2</sub> release killed 1,700 people). SO<sub>2</sub> produces acid rain
- **Tsunamis:** Triggered by volcanic collapse (e.g., Krakatoa 1883 generated 30 m waves, killing 36,000) or submarine eruptions (Hunga Tonga, 2022)
- **Volcanic winter:** Major eruptions inject SO<sub>2</sub> into the stratosphere, forming sulphate aerosols that reflect incoming solar radiation and cool global temperatures. The 1815 Tambora eruption (VEI 7) caused the "Year Without a Summer" in 1816, with crop failures across Europe and North America

## The Volcanic Explosivity Index (VEI)

| VEI | Ejecta Volume                | Classification          | Frequency      | Example                                    |
|-----|------------------------------|-------------------------|----------------|--------------------------------------------|
| 0–1 | <1 million m <sup>3</sup>    | Hawaiian / Strombolian  | Daily          | Kilauea (continuous)                       |
| 2–3 | 1–100 million m <sup>3</sup> | Vulcanian / Sub-Plinian | Monthly–yearly | Nevado del Ruiz (1985, VEI 3)              |
| 4–5 | 0.1–10 km <sup>3</sup>       | Plinian                 | Decades        | Mt St Helens (1980, VEI 5)                 |
| 6   | 10–100 km <sup>3</sup>       | Ultra-Plinian           | Centuries      | Pinatubo (1991, VEI 6)                     |
| 7–8 | 100–1000+ km <sup>3</sup>    | Supervolcanic           | Millennia+     | Tambora (1815, VEI 7); Yellowstone (VEI 8) |

### Case Study: Mount Pinatubo, Philippines (1991) — Volcanic Eruption in an NEE

One of the largest eruptions of the 20th century (VEI 6). Located on Luzon island, ~100 km NW of Manila.

- **Eruption details:** Erupted on 15 June 1991 after 600 years of dormancy. Ejected ~10 km<sup>3</sup> of material; ash column reached 34 km into the stratosphere. 20 million tonnes of SO<sub>2</sub> injected

into the stratosphere

- **Primary effects:** Pyroclastic flows covered ~400 km<sup>2</sup>; ashfall reached 20 cm depth within 40 km, collapsing roofs (the primary cause of death); lahars buried towns and farmland
- **Secondary effects:** 847 deaths (mostly from roof collapse under rain-soaked ash); over 1 million displaced; 8,000 homes destroyed; farmland destroyed for years; \$700 million in property damage. Global cooling of ~0.5°C for 2 years due to stratospheric SO<sub>2</sub> aerosols
- **Prediction and response:** USGS and PHIVOLCS (Philippine Institute of Volcanology) monitored seismometers, tiltmeters, and SO<sub>2</sub> gas emissions. Successfully predicted the eruption and evacuated 58,000 people from a 30 km exclusion zone, saving an estimated 5,000–20,000 lives. Clark Air Base (US military) evacuated.
- **Complicating factor:** Typhoon Yunya struck simultaneously, saturating ash deposits and triggering devastating lahars. This demonstrates the concept of compound hazards
- **Long-term impacts:** Lahars continued for 10+ years during monsoon seasons, burying villages repeatedly; ~20,000 Aeta indigenous people permanently displaced from ancestral lands on the volcano's slopes, losing their cultural identity and livelihoods

**Exam Tip:** Pinatubo is a strong example of successful prediction and evacuation, but also shows the limits of mitigation — secondary hazards (lahars) caused damage for years, and the most marginalised group (Aeta people) suffered the worst long-term impacts. Use it to evaluate how governance and inequality influence vulnerability.

## Seismic Hazards (Earthquakes)

**Earthquake:** A sudden release of stored elastic strain energy in the Earth's crust that creates seismic waves. Caused by tectonic plate movement (most common), volcanic activity, or human activity (induced seismicity from fracking, reservoir loading, mining).

**Focus (hypocentre):** The point underground where the rupture begins and energy is released. Shallow focus (<70 km) earthquakes are generally most destructive.

**Epicentre:** The point on the Earth's surface directly above the focus.

**Seismic waves:** P-waves (primary, compressional, fastest), S-waves (secondary, shear), and surface waves (Love and Rayleigh waves — cause the most damage).

## Measuring Earthquakes

| Scale                             | What It Measures                                       | Key Detail                                                                                                                                       |
|-----------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Moment magnitude ( $M_w$ )        | Total energy released (has replaced the Richter scale) | Logarithmic: each whole number = 10x amplitude increase, ~32x energy. An M7 releases 32x more energy than M6, and ~1000x more than M5            |
| Modified Mercalli Intensity (MMI) | Observed damage and felt effects at the surface        | Qualitative, I–XII scale. Varies by location for the same earthquake (closer to epicentre = higher intensity). Useful for assessing human impact |

## Earthquake Hazards

- **Ground shaking:** Primary hazard; intensity depends on magnitude, focal depth (shallower = more intense surface shaking), distance from epicentre, local geology (soft sediments amplify shaking — "site effects"), and building quality
- **Liquefaction:** Saturated, loose, unconsolidated soil (sand, silt) behaves like liquid when shaken. Buildings sink, tilt, or collapse. Seen extensively in Christchurch 2011 (NZ) where 51% of the central city was damaged
- **Landslides and rock falls:** Shaking destabilises slopes, especially saturated or deforested ones. The 2005 Kashmir earthquake triggered over 2,000 landslides, blocking roads and isolating communities
- **Tsunamis:** Generated when submarine earthquakes (usually M7+, shallow focus, on subduction zones) vertically displace the sea floor. Waves travel across open ocean at 500–800 km/h; amplify in shallow coastal waters to 10–40 m heights
- **Secondary hazards:** Fires from ruptured gas mains (1906 San Francisco fire destroyed 80% of the city), flooding from broken dams or levees, disease outbreaks from contaminated water and disrupted sanitation, economic disruption

### Case Study: Nepal Earthquake (2015) — LIC

- **Magnitude:** 7.8  $M_w$ ; shallow focus (15 km); epicentre 80 km NW of Kathmandu in Gorkha district
- **Cause:** Indian Plate thrusting (underthrusting) beneath Eurasian Plate along the Main Himalayan Thrust; a locked section of the fault ruptured over ~150 km
- **Primary effects:** ~9,000 deaths; 22,000 injured; 604,000 homes destroyed; 288,000 damaged. Historic UNESCO World Heritage monuments in Kathmandu (Dharahara tower, Durbar Squares) collapsed
- **Secondary effects:** Avalanche on Everest killed 22 climbers (deadliest day in Everest history); over 5,000 landslides blocked roads; 3.5 million made homeless; waterborne disease outbreaks in camps; 950,000 children's education disrupted; \$7 billion total damage (~33% of Nepal's GDP)
- **Response:** International aid pledged (\$4.1 billion); India and China sent search and rescue teams within 24 hours. But: remote mountain villages took 5–7 days to reach; helicopter access limited; only one international airport (bottleneck). Corruption and bureaucracy in National Reconstruction Authority slowed rebuilding — 3 years later, ~60% of damaged homes still not rebuilt
- **Vulnerability factors:** Low income (\$730 GNI per capita); unreinforced masonry construction (stone and brick with mud mortar); limited emergency services; mountainous terrain isolating communities; high population density in Kathmandu Valley (built on ancient lake sediments prone to amplification and liquefaction)
- **Aftershock:** M7.3 aftershock on 12 May 2015 caused additional destruction and panic, hampering recovery

### Case Study: Japan Earthquake & Tsunami (2011) — HIC

- **Magnitude:** 9.1  $M_w$  (megathrust at the Japan Trench subduction zone); shallow focus (~30 km); epicentre 70 km offshore of Honshu
- **Earthquake:** Exceptionally long duration of shaking (~6 minutes); moved Honshu 2.4 m east; shifted Earth's axis by 10–25 cm; shortened the day by 1.8 microseconds
- **Tsunami:** Waves up to 40.5 m high (Miyako) struck the Tohoku coast within 30 minutes. Run-up reached 10 km inland in some flat areas. The Sendai Plain was inundated by 5+ km

- **Impacts:** ~15,900 deaths (93% from drowning in tsunami); 2,500 missing; 340,000 displaced; 127,000 buildings totally collapsed; Fukushima Daiichi nuclear plant suffered three meltdowns after the tsunami disabled cooling systems, leading to radiation release, 154,000 evacuations, and a 20 km exclusion zone still partly in force. Total damage: ~\$235 billion (most expensive natural disaster ever)
- **Response:** Japan Meteorological Agency issued a tsunami warning within 3 minutes of the earthquake. Earthquake-resistant buildings performed well — very few collapsed from shaking alone. Self-Defence Forces deployed 107,000 personnel. \$300+ billion reconstruction programme launched
- **Mitigation that worked:** Strict building codes (base isolators, cross-bracing); annual earthquake drills (1 September, anniversary of 1923 Tokyo earthquake); early warning system gave 8–30 seconds notice; tsunami evacuation routes were marked
- **Why impacts were still devastating:** Extreme magnitude exceeded design parameters; 10 m sea walls were overtopped by 14+ m waves; nuclear plant safety measures based on a maximum 5.7 m tsunami assumption; low-lying coastal geography; ageing population in rural Tohoku (limited mobility for evacuation)

**Exam Tip:** When comparing earthquake impacts in HICs vs LICs, use the PEEL structure: Point (state the difference), Evidence (specific data from case studies), Explain (why the difference exists — building codes, wealth, governance, infrastructure), Link (back to the question). Key comparison factors: building standards, emergency services, warning systems, infrastructure quality, governance and corruption, population density, economic resilience, and capacity to recover. Japan 2011 had fewer deaths per unit of magnitude than Nepal 2015 because of superior infrastructure, but the nuclear disaster shows that no amount of preparation fully eliminates risk from extreme events.

## Tropical Storms

**Tropical storm (hurricane/cyclone/typhoon):** An intense low-pressure weather system with sustained wind speeds exceeding 119 km/h (74 mph), forming over warm tropical oceans. Named differently by region: hurricanes (Atlantic/NE Pacific), typhoons (NW Pacific), cyclones (Indian

Ocean/S Pacific). They are heat engines powered by the latent heat released when warm moist air condenses.

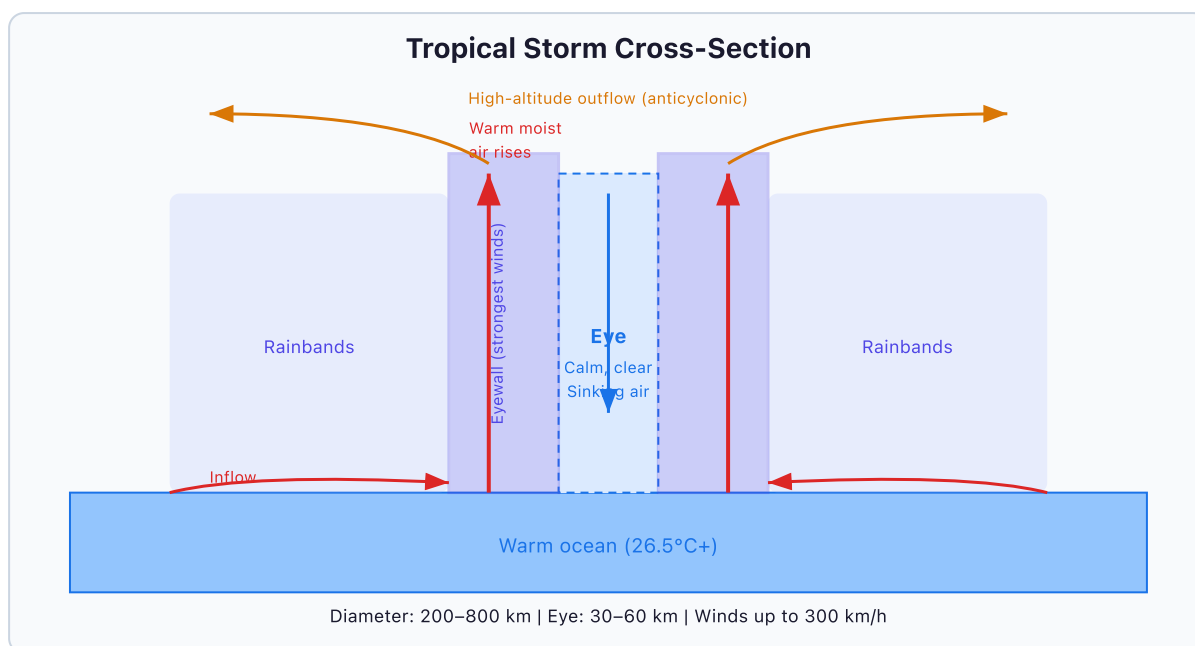
## Formation Conditions

All six conditions must be met simultaneously:

- Sea surface temperature (SST) of at least 26.5°C to a depth of ~60 m (provides energy through evaporation)
- Located at least 5° north or south of the equator (Coriolis effect needed to initiate spin — storms cannot form on the equator itself)
- Low vertical wind shear (consistent wind speeds at different altitudes; high shear disrupts the storm structure)
- Atmospheric instability (warm moist air rises rapidly due to temperature difference with upper atmosphere)
- Pre-existing low-pressure disturbance (e.g., tropical wave, ITCZ trough)
- Sufficient moisture in the mid-troposphere (above 500 hPa level)

**Climate change connection:** Rising SSTs are expanding the area and lengthening the season in which conditions for tropical storm formation are met. While the total number of storms may not increase, evidence suggests storms are becoming more intense (higher peak winds, heavier rainfall), moving more slowly (increasing flood risk), and rapidly intensifying more frequently. Warmer air holds more moisture (~7% more per 1°C of warming), increasing rainfall totals.

## Structure and Mechanics



## Saffir-Simpson Scale

| Category  | Wind Speed (km/h) | Storm Surge (m) | Damage                                                                 |
|-----------|-------------------|-----------------|------------------------------------------------------------------------|
| 1         | 119–153           | 1.2–1.5         | Minimal: some roof and tree damage                                     |
| 2         | 154–177           | 1.8–2.4         | Moderate: significant roof damage; some trees uprooted                 |
| 3 (major) | 178–208           | 2.7–3.7         | Extensive: structural damage to small buildings; flooding inland       |
| 4         | 209–251           | 4.0–5.5         | Extreme: severe structural damage; areas uninhabitable for weeks       |
| 5         | 252+              | 5.5+            | Catastrophic: total destruction of buildings; mass evacuation required |

## Tropical Storm Hazards

- **Storm surge:** A dome of water pushed ashore by the combination of low atmospheric pressure (raises sea level) and onshore winds. The deadliest tropical storm hazard — responsible for ~90% of

hurricane fatalities historically. Bangladesh is extremely vulnerable: Cyclone Bhola (1970) killed 300,000–500,000 people, mainly from storm surge flooding the low-lying delta

- **High winds:** Destroy buildings, infrastructure, power lines, trees. Flying debris is a major secondary hazard. Wind damage increases exponentially: doubling wind speed quadruples the force
- **Intense rainfall & flooding:** Tropical storms can dump 500+ mm of rain in 24 hours. Inland flooding kills more people in the USA than any other hurricane hazard. Hurricane Harvey (2017) dropped 1,539 mm on Houston over 4 days — the highest rainfall from a single tropical system in US history
- **Landslides:** Saturated slopes collapse, especially in mountainous tropical regions. Hurricane Mitch (1998) killed ~11,000 in Central America, mostly from landslides and mudslides
- **Tornadoes:** Can be spawned within the outer rainbands, particularly in the right-front quadrant of the storm

### Case Study: Typhoon Haiyan (Yolanda), Philippines (2013) — NEE/LIC

- **Intensity:** Category 5; sustained winds of 315 km/h with gusts to 380 km/h; one of the strongest tropical cyclones ever recorded at landfall
- **Storm surge:** Up to 5–7 m in Tacloban (population 240,000), devastating the low-lying coastal city. PAGASA warned of storm surges but used the English term, which many residents did not understand; many confused it with normal flooding and stayed in homes
- **Impacts:** Over 6,300 deaths; 28,000 injured; 4 million displaced; 1.1 million homes damaged or destroyed (550,000 totally destroyed); total damage ~\$2.9 billion; Tacloban airport destroyed; 33 million coconut trees destroyed (major livelihood); 90% of Tacloban's structures damaged
- **Why so severe:** Extreme intensity; low-lying coastal settlements built on vulnerable shorelines; high poverty rates (40% below poverty line in Eastern Visayas); inadequate communication of storm surge risk; informal settlements had no resilience; high population density; funnel-shaped Leyte Gulf concentrated the storm surge
- **Response:** Philippines government initially overwhelmed; State of National Calamity declared. International aid from UN, USA (aircraft carrier USS George Washington), UK (HMS Illustrious). \$1.86 billion in international pledges. "Build Back Better" reconstruction programme emphasised elevated housing, mangrove restoration (natural coastal defence), improved early warning systems, and community evacuation plans
- **Lessons:** Communication of risk must be in local languages and concepts; evacuation infrastructure (roads, shelters) must be in place before the event; poverty reduces resilience;

building design standards must account for storm surge, not just wind

### Case Study: Hurricane Katrina, USA (2005) — HIC

- **Intensity:** Category 3 at landfall (had been Category 5 over the Gulf of Mexico); storm surge up to 8.5 m along Mississippi coast
- **Impacts:** 1,836 deaths; 80% of New Orleans flooded after levees failed (designed for Category 3 but inadequately maintained); 1.2 million evacuated; \$125 billion damage; 300,000 homes destroyed; oil refineries shut down, causing national fuel price increases
- **Why an HIC suffered so badly:** New Orleans lies below sea level (bowl-shaped, relying entirely on levees and pumps); levee design flaws (Army Corps of Engineers later admitted failures); slow government response (FEMA delayed 5 days); poverty and racial inequality — 67% of New Orleans was African American, many in low-income areas with no transport to evacuate; 100,000 residents lacked vehicles
- **Recovery:** Still ongoing 10+ years later; population of New Orleans fell from 484,000 to 230,000 and has only partially recovered; \$120 billion in federal recovery spending; \$14.5 billion Hurricane and Storm Damage Risk Reduction System (new levees and floodwalls) completed 2018
- **Lesson:** Even HICs are vulnerable when infrastructure is neglected, inequality is high, and governance fails. Katrina exposed the intersection of hazard risk, poverty, and race in the USA

**Exam Tip:** Use Haiyan and Katrina together to challenge the simple "HICs cope better than LICs" narrative. Katrina shows that wealth alone does not prevent disaster — governance quality, infrastructure investment, and social inequality are critical. The strongest answers discuss how vulnerability is socially constructed, not just a function of GDP.

## Wildfire

**Wildfire:** An uncontrolled fire that burns through vegetation (forest, grassland, scrubland) in rural or wilderness areas. Wildfires require the fire triangle: fuel (vegetation), heat (ignition source), and

oxygen (wind). They are a natural ecological process in many biomes but become a hazard when they threaten human settlements, infrastructure, and lives.

## Conditions for Wildfire

- **Fuel:** Dry vegetation, leaf litter, deadwood, bark. Fuel load (quantity) and fuel moisture (dryness) are critical. Fire-adapted ecosystems (e.g., Australian eucalyptus, Californian chaparral, Mediterranean maquis) have high fuel loads with volatile oils
- **Heat/ignition:** Lightning (natural, causes ~15% of fires but ~50% of area burned), human activity (arson, campfires, discarded cigarettes, power lines, agricultural burning — cause ~85% of ignitions), or spontaneous combustion in extreme heat
- **Oxygen:** Wind increases oxygen supply, drives the fire front, carries embers ahead (spotting), and makes fires unpredictable. Foehn winds (e.g., Santa Ana winds in California, Diablo winds in Northern California) bring hot, dry, gusty conditions that dramatically increase fire risk
- **Weather:** Hot, dry conditions with low relative humidity (<20%); prolonged drought; high temperatures; strong winds. Heatwaves are a major trigger
- **Topography:** Fires spread faster uphill (preheating of fuel above by convection and radiation); steep slopes accelerate fire spread dramatically (doubling slope angle can quadruple spread rate). Canyons and valleys can channel fire and create chimney effects

## Climate Change and Wildfire

Climate change is increasing wildfire risk through multiple pathways: higher temperatures, longer and more intense dry seasons, reduced soil moisture and snowpack, earlier spring snowmelt (extending the fire season), drought stress weakening and killing trees (increasing fuel load), bark beetle outbreaks in warmer winters (killing trees and adding fuel), and increased lightning frequency in some regions. This creates a positive feedback loop: fires release stored carbon as CO<sub>2</sub> (contributing ~5–8% of global emissions in severe fire years), which enhances warming, which increases fire risk. The global area burned has increased significantly since 2000, with the fire season lengthening by 18.7% globally between 1979 and 2013.

## Wildland-Urban Interface (WUI)

The WUI is the zone where human development meets or intermingles with wildland vegetation. As populations expand into forested and semi-rural areas (driven by housing demand and quality of life

preferences), more people and structures are exposed to wildfire risk. In the USA, 46 million homes are in the WUI. In Australia, the "bush-urban interface" accounts for most fire casualties. The WUI is the primary reason wildfires have become more costly and deadly despite improved firefighting technology.

### Case Study: Australian Bushfires, "Black Summer" (2019–2020)

- **Scale:** Over 18.6 million hectares burned across New South Wales, Victoria, Queensland, South Australia, and Western Australia — an area larger than England and Wales combined. The fires burned from September 2019 to March 2020
- **Causes:** Record-breaking drought (worst in 800+ years based on dendrochronology); 2019 was Australia's hottest and driest year on record; extreme heat (Penrith, western Sydney, reached 48.9°C on 4 January 2020); strong, erratic winds; some human ignition (dry lightning caused most large fires); climate change identified as a key driver by the Bureau of Meteorology and CSIRO
- **Human impacts:** 34 direct deaths (including 9 firefighters); 445 deaths attributed to smoke inhalation complications; 5,900+ buildings destroyed including 2,779 homes; \$10 billion AUD (\$7 billion USD) economic losses; smoke blanketed Sydney for weeks (PM2.5 levels reached 11x the hazardous threshold — equivalent to smoking 37 cigarettes per day); smoke reached New Zealand, South America, and circled the globe
- **Ecological impact:** Approximately 3 billion animals killed or displaced; 143 million mammals, 2.46 billion reptiles, 180 million birds, 51 million frogs affected. Habitat destruction for endangered species (koalas: estimated 60,000 affected; 30% of koala habitat in NSW destroyed). Some fire-sensitive ecosystems (Gondwanan rainforest in NSW, ancient Huon pines in Tasmania) may not recover as fire intensity exceeded their evolutionary tolerance
- **Response:** Australian Defence Force deployed (Operation Bushfire Assist — 6,500 military personnel, largest peacetime deployment); international firefighters from USA, Canada, NZ; A\$2 billion National Bushfire Recovery Fund; Royal Commission into National Natural Disaster Arrangements recommended national coordination improvements
- **Climate link:** The Bureau of Meteorology and CSIRO concluded that climate change increased fire risk by extending fire seasons, increasing drought severity, and raising temperatures. The fires contributed 830 million tonnes of CO<sub>2</sub> — more than double Australia's normal annual emissions, demonstrating the fire-climate feedback loop

### Case Study: California Wildfires — Camp Fire, Paradise (2018)

- **Scale:** The Camp Fire burned 62,000 hectares and was the deadliest and most destructive wildfire in California's history
- **Deaths:** 85 people killed, mostly elderly residents unable to evacuate. Paradise (population 27,000) was effectively destroyed — 18,804 structures burned, including 13,972 homes
- **Cause:** A faulty Pacific Gas & Electric (PG&E) transmission line ignited dry vegetation. PG&E was found criminally liable and filed for bankruptcy (\$30+ billion in fire-related liabilities across multiple fires)
- **Speed:** The fire moved so fast (driven by Jarbo winds at 80+ km/h) that it consumed the equivalent of 80 football fields per minute, giving residents minutes rather than hours to evacuate. Only one main road out of Paradise created a deadly bottleneck
- **Context:** California's fire season has lengthened by 75 days since the 1970s. The WUI expansion means millions now live in high-fire-risk zones. The state now mandates home hardening, defensible space, and improved evacuation planning

## Hazard Risk and Vulnerability

**Risk:** The probability of a hazard occurring AND causing harm. Risk depends on the interaction between the hazard event, the exposure of people and assets, and their vulnerability.

**Vulnerability:** The susceptibility of a population to harm from a hazard, determined by economic, social, political, and environmental factors.

**Resilience:** The ability of a community to anticipate, prepare for, respond to, and recover from a hazard event. High resilience reduces actual impact relative to potential impact.

$$\text{Risk} = \text{Hazard} \times \text{Exposure} \times \text{Vulnerability} \div \text{Capacity (to cope)}$$

### Factors Affecting Vulnerability

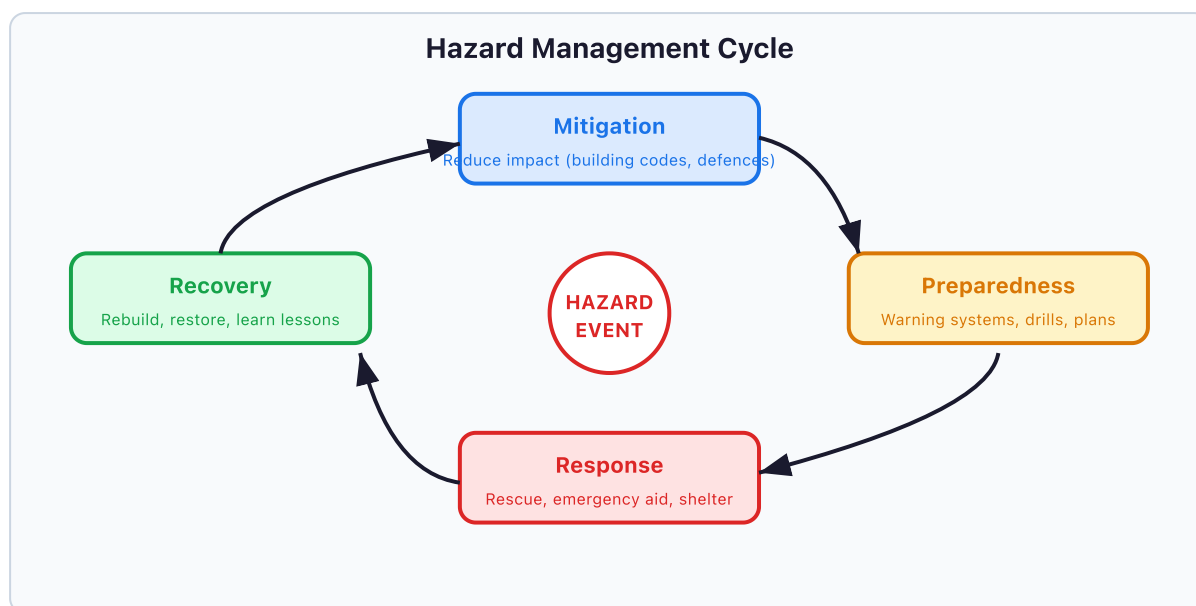
| Factor | Increases Vulnerability | Reduces Vulnerability |
|--------|-------------------------|-----------------------|
|--------|-------------------------|-----------------------|

|                    |                                                                     |                                                                                                     |
|--------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Wealth/development | Poverty; no insurance; no savings; subsistence dependence           | Resources for building codes, emergency services, insurance systems                                 |
| Governance         | Corruption; weak institutions; no enforcement of regulations        | Strong planning laws; transparent governance; enforcement of building codes                         |
| Education          | Low hazard awareness; inability to interpret warnings               | Disaster preparedness education; community drills; local knowledge                                  |
| Technology         | No monitoring or warning systems; poor communication networks       | Seismometers, satellite monitoring, early warning systems, mobile phone alerts                      |
| Infrastructure     | Poor building standards; congested roads; no evacuation shelters    | Earthquake-resistant buildings; designated evacuation routes and shelters; redundant infrastructure |
| Demographics       | High density in hazard zones; elderly/young populations; disability | Dispersed population; able-bodied, mobile population; low dependency ratio                          |
| Social capital     | Isolated individuals; weak community networks; social inequality    | Strong community bonds; mutual aid networks; social cohesion                                        |

**Exam Tip:** The concept of vulnerability is central to OCR's approach to hazards. The examiner wants you to move beyond describing the physical event to explaining why some people are more vulnerable than others. Vulnerability is socially constructed — it is shaped by inequality, governance, and power. Link to the idea that "there is no such thing as a natural disaster" (only natural hazards that become disasters when they interact with vulnerable populations).

## Hazard Management Strategies

### The Hazard Management Cycle



## Prediction and Monitoring

| Hazard          | Prediction Methods                                                                                                                                                                                                                                                             | Accuracy                                                                                                                                                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Volcanoes       | Seismometers (earthquake swarms below the volcano); tiltmeters and GPS (ground deformation from magma movement); gas monitoring (SO <sub>2</sub> increases as magma nears surface); satellite thermal imaging; historical records and eruption frequency analysis              | Moderate–good for weeks/days before eruption. Pinatubo 1991 is a success story. But: false alarms are common, and some eruptions occur with little warning |
| Earthquakes     | <b>Cannot be reliably predicted.</b> Seismic gap theory (areas overdue for a rupture); historical recurrence intervals; palaeoseismology (studying ancient earthquake deposits); stress transfer modelling. Animal behaviour changes are anecdotal and unproven scientifically | Low. No method can predict the exact time, location, and magnitude. Focus must be on preparedness rather than prediction                                   |
| Tropical storms | Satellite tracking (geostationary and polar-orbiting); weather buoys and dropsondes; radar; numerical weather prediction (NWP) computer models. Track forecasts: 3–5 day predictions are now highly accurate (average track error ~200 km at 72 hours)                         | Good for track prediction. Intensity forecasting is weaker, especially for rapid intensification events                                                    |

|           |                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tsunamis  | Pacific Tsunami Warning Centre (PTWC); deep-ocean pressure sensors (DART buoys detect passing waves in open ocean); seismometer networks trigger warnings within minutes of an earthquake. Indian Ocean Tsunami Warning System established after 2004 disaster | Good once an earthquake occurs. Warning time depends on distance from source (minutes for near-field, hours for far-field). The 2004 Indian Ocean tsunami had no warning system in place, causing 230,000 deaths |
| Wildfires | Fire weather indices (combining temperature, humidity, wind, drought); satellite hot-spot detection (MODIS, VIIRS); lightning detection networks; fuel moisture monitoring; fire behaviour modelling                                                           | Moderate for broad risk assessment. Specific fire prediction (where and when ignition occurs) is impossible. Satellite detection allows rapid response once fire starts                                          |

## Protection and Adaptation

- **Earthquake-resistant buildings:** Steel-reinforced frames, cross-bracing, base isolators (rubber pads that absorb seismic energy), deep pile foundations in soft soils, tuned mass dampers (e.g., Taipei 101's 660-tonne pendulum), shear walls, flexible utility connections. Japan's building codes are the world's strictest; buildings sway but do not collapse
- **Land-use planning:** Restricting development in high-risk zones (floodplains, volcanic flanks, liquefaction-prone areas, active fault zones, steep fire-prone slopes). Hazard zonation maps guide planning decisions. Politically difficult when land is valuable or communities already exist in risk zones
- **Structural defences:** Sea walls and levees (protect against storm surges and tsunamis but can be overtopped or create false security); breakwaters; coastal mangrove restoration (natural buffer — reduces tsunami wave energy by up to 70%); fire breaks; flood barriers
- **Education and drills:** Japan's annual earthquake drills (Disaster Prevention Day, 1 September); "Drop, Cover, Hold On" earthquake drill; school-based tsunami education (Kamaishi "miracle" — 3,000 school children evacuated successfully during 2011 tsunami because of regular drills); community fire safety programmes
- **Community-based disaster risk reduction (CBDRR):** Local communities participate in risk assessment, evacuation planning, and building design. Combines scientific knowledge with indigenous/local knowledge. Emphasises empowerment of vulnerable groups (women, elderly, disabled). Increasingly recognised as essential alongside top-down approaches

- **Insurance and financial preparedness:** National insurance schemes (USA's National Flood Insurance Programme); catastrophe bonds; international disaster relief funds. Insurance enables faster recovery but is often unaffordable or unavailable in LICs

**Exam Tip:** For hazard management questions, use the Park Model (shows response and recovery trajectory over time — quality of life drops at disaster, then recovers; the speed and level of recovery depends on preparedness and resources) to structure your evaluation. Discuss short-term responses (search and rescue, emergency shelters, aid) vs long-term mitigation (building codes, land-use planning, warning systems). Always evaluate whether strategies are realistic for the level of development — a building code is meaningless if people cannot afford to comply, and an early warning system fails if there are no evacuation routes. The strongest answers consider the political economy of disaster risk: who benefits from current arrangements, and who is left exposed.

## Why People Live in Hazardous Areas

- **Fertile volcanic soils:** Volcanic ash weathers to produce nutrient-rich soils (e.g., Java, Indonesia supports 150 million people on an area the size of England; rice yields are among the world's highest)
- **Geothermal energy:** Iceland generates ~25% of its electricity from geothermal sources; Kenya and the Philippines also benefit from volcanic geothermal power
- **Tourism and economic opportunity:** Volcanic landscapes attract tourism (Mount Fuji: 300,000 climbers/year; Vesuvius and Pompeii: 2.5 million visitors/year). Coastal areas prone to tropical storms are also economically vital (fishing, trade, tourism)
- **Poverty and lack of alternatives:** The poorest communities often occupy the most hazard-prone land (floodplains, unstable slopes, volcanic flanks) because it is the only affordable or available land. Poverty traps people in risky locations
- **Optimism bias and risk perception:** People underestimate the probability of rare events occurring in their lifetime. "It hasn't happened here before" mentality
- **Historical and cultural ties:** Deep attachments to ancestral lands; cultural identity linked to place; reluctance to relocate
- **Lack of hazard awareness or education:** In some areas, the time interval between major events exceeds living memory, so awareness of risk is low

- **Infrastructure lock-in:** Once cities and major infrastructure are built in hazardous locations, relocation is prohibitively expensive (e.g., Tokyo, San Francisco, Manila)

**Exam Tip:** This topic often appears as an essay question: "Assess the view that the impact of tectonic hazards is determined more by human factors than physical factors." Structure your answer by discussing physical factors (magnitude, depth, type, location, secondary hazards) vs human factors (development level, building quality, governance, population density, vulnerability, warning systems). Use your case studies (Nepal vs Japan) to illustrate. Conclude that while the physical event creates the hazard, it is human factors — especially inequality and governance — that determine whether it becomes a disaster.